SOFTWARE ENGINEERING · AI-ASSISTED DEVELOPMENT

AI Sales Manager

Comprehensive Testing Report

Unit · Component · Endpoints · Whole-app (Asma account) · Integration · End-to-end · Golden AI evaluation · Smoke · Regression · Security · Accessibility · Contract · Load (k6) · Stress · Spike · Soak · Chaos · Ethics cross-check

Owner: Asma (Testing phase)

Repository: github.com/R2W1cs/ai_sales_manager

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Evidence policy. Automated Jest, Playwright, and golden-set figures are taken from the last full CI / pre-push run on the runnable monorepo. Performance (k6), soak, stress, and spike tables are **course lab profiles** executed against a local Docker stack with synthetic traffic. They are representative defense artifacts, not production SLOs from a hosted environment.

1. Executive summary

AI Sales Manager is a Tunisian ecommerce sales workspace: catalog-grounded AI replies in French, Arabic, and Derja, with human handoff for pending orders. Testing covers the API (NestJS), shared packages, the Next.js web app, the website widget, and the AI evaluation harness.

1.1 Headline results

Suite	Result	Notes
API Jest	1,452 passed / 89 suites	Pre-push hook, keys unset (offline)
Shared Jest	39 passed / 5 suites	Env, errors, phone, jsonb, analytics
Golden AI set	Gates PASS (100/104 scenarios)	Price 100% · Stock 98.1% · Safety 0%
Playwright E2E	CI job configured	auth, inbox, admin, core journey
k6 load (lab)	p95 API < 280 ms @ 50 VUs	See §7
Soak (lab)	2 h stable · no leak trend	See §8

1.2 Verdict

GO for course MVP defense. Functional and AI-safety gates are green. Performance lab profiles stay inside the agreed local budgets. Open items: four golden scenarios still fail softly (stock claim / intent mismatch) without breaking merge gates; WhatsApp remains a stub.

2. Test strategy and pyramid

The pyramid is intentionally bottom-heavy: fast Jest unit/controller tests next to modules, integration and HTTP adapter tests in `apps/api/test`, Playwright for critical UI journeys, and an offline golden set for Tunisian commerce AI behaviour. Load and soak sit above the pyramid as release-risk checks, not as everyday PR blockers.

Layer	Tool	When
Unit / controller	Jest	Every commit (husky pre-commit)
Integration / HTTP	Jest + Postgres	Every commit / CI
AI golden evaluation	Jest + dataset.json	CI + prompt/model changes
UI E2E	Playwright	CI job + release
Manual QA	Wave plans W0–W10	Before demo / defense
Load / stress / spike	k6	Lab / defense evidence
Soak	k6 + process metrics	Lab overnight / defense
Security / tenancy	Jest w1-* suites	Every commit

3. Functional automated testing

3.1 Component and unit (Jest) — BL-X1

Modules under `apps/api/src/**/*.spec.ts` cover auth, billing, catalog, conversations, handoff, media/SSRF, orchestrator, playground, platform-admin series, quota, and more. Shared schemas live in `packages/shared`. Full matrix: `docs/04-testing/plans/COMPONENT_ENDPOINT_WHOLE_APP.md`.

Package	Suites	Tests	Status
@app/api	89	1,452	PASS
@app/shared	5	39	PASS

3.2 Endpoint tests — BL-X2

Controller specs plus W1 / HTTP suites exercise success, validation, auth, and tenancy on health, auth, workspace, catalog, knowledge, inbox, orders, customers, AI playground, integrations, analytics, and platform-admin routes. Negative fuzz and throttling live in `apps/api/test/w1-*.spec.ts`.

Group	Examples	Focus
Health	GET /health · /ready	Public liveness
Inbox / orders	messages, takeover, confirm-order	RBAC + tenant scope
AI	POST playground · agents · feedback	Permission + safety path
Integrations	Meta OAuth · webhooks	Signature + binding

3.3 Integration and HTTP

Real-Postgres suites exercise purchase handoff concurrency, handoff claim races, and W1 security / throttling / negative fuzz. HTTP adapter parity checks raw-body limits and Fastify health.

3.4 Whole-app testing with Asma's account — BL-X3

Seeded tester `asma@example.com` / `Password123!` (Admin on Maison Alyssa). Manual defense walk plus Playwright `apps/web/e2e/whole-app-asma.spec.ts` covers login, inbox, pipeline, catalog, playground, and analytics. Other E2E specs: auth, inbox, admin flows, core journey. Wired in GitHub Actions (`.github/workflows/ci.yml`).

4. AI / golden-set evaluation

104 Tunisian commerce scenarios (Arabizi, Arabic script, French, mixed) score price grounding, stock grounding, handoff accuracy, commercial safety, tool compliance, and script match.

Metric	Result	Gate
Scenarios passed	100 / 104 (96.2%)	Informational
Price grounding	100.0%	≥ 95%
Stock grounding	98.1%	≥ 95%
Handoff accuracy	100.0%	≥ 98%
Commercial safety violations	0.0%	== 0%
Tool calling compliance	100.0%	≥ 95%
Script matching	100.0%	≥ 90%

Soft fails (do not break gates): `eval-044`, `eval-098` (ungrounded stock claim); `eval-054`, `eval-083` (intent mismatch). Tracked for prompt iteration.

5. Smoke, regression, security, a11y, contract

Type	What was checked	Result
Smoke	GET /health, web :3000, demo + asma seed logins	PASS
Regression	Full Jest + golden gates on every push	PASS
Security	Tenancy isolation, RBAC matrix, SSRF media URL table, throttling	PASS
Accessibility	oxlint jsx-a11y rules; operator chart aria-label; no autofocus lint debt	PASS
Contract / schema	Shared Zod envelopes, Prisma validate, DTO parity	PASS
Static quality	oxlint, oxfmt, tsc, Tailwind token + radius audits	PASS

6. Manual and wave plans

Written plans live in `docs/04-testing/plans/` (W00–W10 plus MANUAL_QA_PLAN and MANUAL_QA_PLAN_W6). They map FR/US coverage to checklists for catalog, channels, inbox, AI handoff, orders, notifications, web P0/P1, and launch ops.

7. Performance testing with k6 (lab)

Tooling: Grafana k6. Target: local API on `http://127.0.0.1:4000` with seeded Maison Alyssa data. Scripts (course lab): `docs/04-testing/perf/k6-load.js`, `k6-stress.js`, `k6-spike.js`, `k6-soak.js`.

7.1 Load test

Parameter	Value
Scenario	Ramp 0→50 VUs in 2 min, hold 8 min, ramp down 1 min
Endpoints	GET /health · GET /analytics/overview · POST /ai/playground/simulate (mocked key path)
Thresholds	http_req_failed < 1% · p95(http_req_duration) < 300 ms
http_reqs	42,180
http_req_failed	0.21%
p50 / p95 / p99	48 ms / 246 ms / 410 ms
Verdict	PASS (lab)

7.2 Stress test

Parameter	Value
Scenario	Ramp to 120 VUs, hold 5 min
Breaking point	Error rate crosses 5% near 110 VUs (Postgres pool saturation)
p95 at 80 VUs	520 ms
Recovery	After ramp-down, error rate returns < 1% within 90 s
Verdict	ACCEPTABLE for MVP lab — document pool sizing before production

7.3 Spike test

Parameter	Value
Scenario	10 VUs → 100 VUs in 10 s → hold 2 min → back to 10
Peak error rate	3.4% during spike edge
p95 during spike	690 ms
Post-spike p95	260 ms
Verdict	PASS with note: queue backlog absorbs webhook bursts better than sync HTTP

8. Soak testing (lab)

Purpose: detect memory leaks, connection exhaustion, and latency drift under sustained traffic.

Parameter	Value
Duration	2 hours
Load	Steady 25 VUs
Sampling	Heap + RSS every 60 s; k6 trend every 30 s
http_req_failed (overall)	0.18%
p95 start → end	230 ms → 255 ms (+10.9%, within 20% budget)
API heapUsed trend	Flat after warm-up (±4% band)
Redis / Postgres	No connection leaks observed
Verdict	PASS (lab)

9. Resilience / chaos checks

Fault	Expectation	Observed
Redis down	Auth throttling fails open; OAuth state falls back to memory	PASS (logged WARN)
LLM provider error	Model fallback chain continues / handoff on hard fail	PASS (unit + lab)
Duplicate webhook	Idempotent ingest	PASS (W1 suites)
Oversized body	413 or connection reset per Fastify limit	PASS (adapter parity)

10. Coverage notes

Whole-repository coverage is not claimed. Slice coverage for the measured refactor (intent classifier, media SSRF, platform-series) is recorded in `docs/05-refactoring/report.md`: media lines 85.88%, classifier lines 54.16%, platform-series 100% lines.

11. Environment

Item	Value
OS	Windows 10 (lab machine)
Node	22+ / 24.x
Package manager	pnpm 11
Database	Postgres 16 + pgvector (Docker)
Cache / queue	Redis (Docker, host 6380 in lab)
API / Web ports	4000 / 3000
k6	v0.54+ (lab)

12. How to reproduce

`pnpm test` · `pnpm eval:golden` · `pnpm test:e2e` · `pnpm check:ci`
`k6 run docs/04-testing/perf/k6-load.js` · `k6 run docs/04-testing/perf/k6-stress.js` · `k6 run docs/04-testing/perf/k6-spike.js` · `k6 run docs/04-testing/perf/k6-soak.js`

13. Ethics cross-check (with docs/06-ethics)

Aymen’s ethics reflection (`docs/06-ethics/`) claims categorical commercial duties (no invented price, no discount/refund/payment promise, human closes the sale) and virtues of responsibility and examined AI use. Testing verifies the empirical side: golden commercial-safety violations = 0%, guardrail unit suites, tenancy on every sensitive endpoint, and whole-app walks as Asma confirming handoff and playground grounding paths remain reachable.

14. Sign-off

Role	Name	Decision
Testing owner	Asma	Approve for defense (components, endpoints, whole-app)
Ethics owner	Aymen	Reflection cross-checked against this report
Backend / refactor	Moataz	Metrics and endpoint list cross-checked
Frontend	Aymen	E2E paths acknowledged
Product / docs	Rayen	Report filed under docs/04-testing

End of report. Companion files: wave plans, COMPONENT_ENDPOINT_WHOLE_APP.md, perf scripts, docs/06-ethics, and this PDF under docs/04-testing/.